

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Condensed Matter Physics Lab
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L-T-P: 0-0-3

Credits:

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

1. Determination of permittivity of air
 2. Determination of dielectric constant of a liquid
 3. Measurement of ionic conductivity of a given crystal as a function of temperature and estimation of activation energy
 4. Variation of Hall co-efficient with temperature for given extrinsic semiconductor
 5. Measurement of g factor of electron of a material using ESR signal
 6. Measurement of magnetization of a given sample as a function of field.
 7. Determination of susceptibility of paramagnetic liquid
 8. Study the variation of Hall voltage in metal with magnetic field
 9. Determination of thermal linear expansion co-efficient.
 10. Measurement of low temperature resistivity of metal / semiconductor
 11. Measurement of high temperature resistivity of metal / semiconductor
 12. Measurement of dielectric constant of a given sample as a function of temperature and determine the transition temperature.
 13. Study to junction voltage and determination of energy band gap of a given transistor
 14. Measurement of magnetoresistance for different fields.
 15. Measurement of magnetic susceptibility of a given sample
 16. Determine the spin-lattice relaxation time of given sample using pulsed NMR.
- Study of variation of Hall voltage developed in the semiconductor and determination of Hall co-efficient

Reference Books:

Lab manual